

Training-Free Generation of Protein Sequences from Small Family Alignments via Stochastic Attention

Jeffrey D. Varner^{a,1}

^aR.F. Smith School of Chemical and Biomolecular Engineering, Cornell University, Ithaca, NY 14850

This manuscript was compiled on March 23, 2026

1 Generating novel protein sequences that respect the statistical constraints of a family typically requires training deep generative models on
2 thousands to millions of examples. Yet most protein families are small: the median Pfam family has fewer than 100 members, a regime where
3 learned models overfit or collapse. We propose *stochastic attention* (SA), a training-free sampler that treats the modern Hopfield energy
4 over stored protein sequences as a Boltzmann distribution and draws samples via Langevin dynamics. The score function is given in closed
5 form by the residual of a single softmax attention operation, eliminating the need for a trained score network, external pretraining data, or
6 graphics processing unit (GPU) resources. Across eight Pfam families spanning 37 to 420 sequences and 23 to 262 residues, SA generates
7 sequences with low amino acid composition divergence, substantial novelty, and structural plausibility confirmed by ESMFold and AlphaFold2.
8 Head-to-head comparison with profile HMMs, EvoDiff, and the MSA Transformer shows that SA is the only method to simultaneously achieve
9 low composition divergence, genuine novelty, and sequence identity within the natural within-family range (30–70%); competing methods
10 either drift outside the family or produce near-copies of stored patterns. The critical inverse temperature is predicted from principal component
11 analysis (PCA) dimensionality alone with high accuracy, enabling fully automatic operation from a seed alignment. Independent validation by
12 ESM2-650M and cross-referencing against deep mutational scanning experiments confirm that over 90 percent of SA-generated substitutions
13 correspond to experimentally tolerated mutations.

protein sequence generation | stochastic attention | Hopfield networks | Langevin dynamics | training-free generative models

1 With protein structure prediction largely solved by AlphaFold2 (1) and RoseTTAFold (2), the frontier
2 has shifted toward *designing* novel protein sequences with desired properties (3, 4). Generative
3 approaches now span a broad landscape, from classical profile hidden Markov models (pHMMs) (5, 6) and
4 Potts models (7–9) to deep generative models including variational autoencoders (10, 11), autoregressive
5 language models (12, 13), alignment-processing transformers (14), MSA-conditioned diffusion (15), and
6 structure-conditioned methods such as ProteinMPNN (16) and RFdiffusion (17). Yet all sequence-level
7 approaches share a common requirement: large training corpora. Language models are pretrained on
8 millions of UniRef50 sequences; even MSA-conditioned methods require either extensive pretraining or
9 large input alignments. This reliance on scale creates a fundamental blind spot, because the median Pfam
10 family contains fewer than 100 full-length members (18), over 60% of families have fewer than 50 seed
11 sequences, and thousands have fewer than 20, well below the minimum viable training set for any deep
12 generative architecture. In this scarce-data regime, deep models overfit and collapse in diversity, while
13 pHMMs emit sequences that drift far from the family’s functional envelope. What is needed is an approach
14 that can extract the statistical structure of a protein family from whatever data is available, even a few
15 dozen sequences, without learning parameters that could overfit.

16 A separate line of work provides exactly this capability. Ramsauer et al. (19) showed that a single softmax
17 attention operation is equivalent to one step of the retrieval dynamics on the modern Hopfield energy,
18 building on the dense associative memory framework of Krotov and Hopfield (20). This connection endows
19 any set of stored examples with a well-defined energy landscape: the examples become memory patterns,
20 and the Boltzmann distribution over this landscape assigns high probability to states that resemble the
21 stored set. The energy itself has a transparent structure: a quadratic term $\frac{1}{2}\|\xi\|^2$ pulls the state toward the
22 origin, while a log-sum-exp term $-\frac{1}{\beta}\log\sum_k\exp(\beta\mathbf{m}_k^\top\xi)$ pulls it toward stored memories. The gradient of
23 the log-sum-exp is precisely the softmax-weighted combination of stored patterns, i.e., a single attention
24 operation. The score function of the Boltzmann distribution is therefore the *residual* of attention: the
25 difference between the attention output and the current state. This residual is exact and available in

closed form, so Langevin sampling can proceed without learning any parameters, training a score network, or approximating gradients. We exploit this insight to propose *stochastic attention* (SA), a training-free protein sequence generator that combines the exact Hopfield score function with Langevin dynamics. Our prior work (21) introduced SA and validated it on synthetic data, MNIST, and a single protein family.

In this study, we explored the extent to which stochastic attention can serve as a practical generative model for real protein families. We systematically evaluated SA across eight diverse Pfam families spanning a ten-fold range of family sizes and a seven-fold range of sequence lengths, assessing compositional fidelity, sequence novelty, and structural plausibility using both ESMFold and AlphaFold2. We characterized how generation quality scales with family size, uncovering an empirical law that predicts the critical temperature from alignment statistics alone and eliminates the need for a temperature sweep. We tested robustness in the extreme scarce-data regime down to $K=20$ sequences, compared SA head-to-head against four established baselines (profile HMMs, EvoDiff, MSA Transformer, and a Potts model), and examined whether the generated sequences preserve the pairwise residue covariation that reflects structural contacts and functional couplings. Finally, we sought independent validation of biological plausibility through ESM2-650M pseudo-perplexity scoring and cross-referencing against published deep mutational scanning datasets.

Results

We ran stochastic attention on all eight Pfam families (Table 1) and observed that the method consistently generated novel protein sequences while preserving family-level amino acid statistics, across a ten-fold range of family sizes and a seven-fold range of sequence lengths (Fig. 1; SI Appendix, Table S1). We first confirmed that the sampler converged to the correct Boltzmann distribution by comparing the outputs of the unadjusted Langevin algorithm (ULA) and its Metropolis-adjusted variant (MALA): acceptance rates exceeded 99.6% in every family (SI Appendix, Table S7), confirming negligible discretization bias and justifying the use of the computationally cheaper ULA in all subsequent experiments. Generated sequences were not near-duplicates of stored patterns: zero sequences across all eight families exceeded 90% identity to any stored sequence (SI Appendix, Table S9), and the generated ensemble was insensitive to initialization strategy (SI Appendix, Table S8). In the generation regime (inverse temperature $\beta \approx 2\beta^*$, just above the entropy inflection point where pattern-specific basins first emerge), SA achieved Kullback–Leibler (KL) divergences below 0.06 in every family, with several below 0.02. Novelty ranged from 0.40 (Defensin_beta) to 0.65 (WW), indicating substantial departure from any single stored memory, and mean pairwise cosine distance was ≈ 1.0 in every family, confirming that the sampler produced distinct, well-separated sequences rather than collapsing to a single mode. Nearest sequence identity to a stored pattern was 51–66%, comparable to the typical within-family identity range, indicating that generated sequences remained within the family’s functional envelope while departing meaningfully from stored patterns. By contrast, bootstrap samples had zero novelty by construction and Gaussian perturbation produced negligible novelty (< 0.02), confirming that small additive noise is insufficient to escape the basin of the nearest stored pattern.

Position-specific analysis of the Kunitz domain revealed why SA preserves family statistics (SI Appendix, Fig. S5). SA-generated sequences preserved all 11 positions with greater than 90% conservation in the seed alignment, including all six cysteines forming the three disulfide bonds essential to the Kunitz fold, while introducing 17–25 substitutions concentrated at variable positions. Per-position Shannon entropy correlated strongly between the SA-generated ensemble and the stored alignment ($r=0.968$), confirming that the method correctly identified which positions tolerate variation and which do not. This behavior followed directly from the Boltzmann sampling mechanism: at strongly conserved positions, all stored patterns agreed and the softmax weights concentrated probability on the consensus residue; at variable positions, stored patterns diverged and the sampler drew from the local distribution of tolerated amino acids.

Subsampling the WW domain at $K \in \{20, 50, 100, 200, 400\}$ with five independent replicates per condition characterized how generation quality changed with decreasing family size (SI Appendix, Fig. S2). Stochastic

attention maintained low KL divergence across the entire range (0.019 ± 0.008 at $K=20$, improving to 0.010 ± 0.002 at $K=400$), while the PCA dimensionality grew from $d=17$ – 18 components at $K=20$ to $d=182$ – 183 at $K=400$ and the critical temperature tracked this growth ($\beta^*=2.7$ to 5.5). Despite the roughly ten-fold increase in effective dimensionality, SA generation quality remained stable, suggesting that the entropy inflection criterion successfully adapted the operating temperature to the geometry of the memory matrix at each scale. Novelty increased with K (from 0.47 to 0.74), reflecting the richer landscape of a larger memory set, while nearest sequence identity decreased correspondingly (from 0.71 to 0.56), indicating that the sampler explored further from stored patterns when more patterns were available to collectively shape the energy landscape. Replication across three additional families at $K=20$ (SH3, Kunitz, zf-C2H2) confirmed that this robustness is not family-specific (SI Appendix, Table S2).

The scaling study revealed that β^* tracked PCA dimensionality, prompting us to ask whether this relationship could predict β^* from alignment statistics alone, without a temperature sweep (Fig. 2). We pooled 33 data points (the 8 Pfam families and 25 WW scaling replicates) and found that a simple two-parameter model, $\beta^* \approx 1.57 + 0.28\sqrt{d}$ (bootstrap standard error [SE]: intercept ± 0.07 , slope ± 0.007), explained 97% of the variance ($R^2=0.97$). The intercept of 1.57 recovers the universal softmax inflection baseline, and the fitted slope of 0.28 is roughly half the $\sqrt{d}$ coefficient predicted for random patterns. This reduction arises from two effects: concentration of measure on $\mathbb{S}^{d-1}$ guarantees that cross-similarities between stored patterns scale as $\mathcal{O}(1/\sqrt{d})$, setting the $\sqrt{d}$ dependence, while self-similarity anchoring (each stored pattern has unit inner product with itself, creating a gap of 0.55 – 0.82 over the next-nearest pattern) shifts the inflection to lower β than a Gaussian mean-field model predicts; together these effects reflect the structured similarity spectrum of real protein sequences arising from conserved positions, correlated substitutions, and phylogenetic signal (SI Appendix, Section S18). Adding further predictors (mean column entropy, effective number of sequences, spectral concentration) produced negligible improvement ($\Delta R^2 < 0.001$), indicating that PCA dimensionality captures most of the relevant variation. Refitting on the eight Pfam families alone yielded a nearly identical relationship ($R^2=0.95$), confirming that the result is not driven by overrepresentation of the WW family. This regression provides a practical shortcut: given a new family, one can estimate β^* from PCA alone and set the generation temperature without a sweep.

We compared stochastic attention against three established learned baselines, profile HMM emit, EvoDiff (MSA-conditioned diffusion), and the MSA Transformer (Gibbs masked language model sampling), across all eight families (Fig. 3). Reading the three panels jointly is essential: no single metric tells the full story, and a method that scores well on novelty while failing on compositional fidelity or sequence identity has simply generated sequences that no longer belong to the family. Profile HMM emit and the MSA Transformer achieved high novelty scores, but this apparent advantage is illusory: their nearest sequence identity to stored patterns fell to 11 – 59% in several families, well below the typical within-family range of 30 – 70% (shaded band, Fig. 3C). Sequences below this range have drifted outside the family and are unlikely to retain its function; sequences above it are near-copies of stored patterns with insufficient novelty. The high novelty of HMM emit and the MSA Transformer reflects compositional drift, not meaningful exploration of family sequence space. EvoDiff balanced the metrics more effectively in shorter families but required pretraining on millions of UniRef50 MSAs, scaled poorly with sequence length (~ 14 hours for 150 Pkinase sequences on CPU, though GPU inference would reduce this substantially, versus seconds for SA on the same hardware; SI Appendix, Table S10), and suffered a sharp degradation in compositional fidelity on the longest family (Pkinase KL= 0.20 , an order of magnitude worse than SA).

Stochastic attention was the only method that simultaneously achieved all three desiderata (SI Appendix, Table S12): KL divergence below 0.05 in seven of eight families (a KL of 0.05 over 20 amino acids corresponds to an average absolute frequency error of less than 0.5 percentage points per residue type, well within finite-sample variation), novelty of 0.40 – 0.65 indicating genuine departure from stored patterns (zero generated sequences exceeded 90% identity to any stored pattern, and zero were within two substitutions of a stored sequence across all eight families), and sequence identity of 51 – 66% firmly within the natural within-family range. This combination (compositionally faithful, genuinely novel, and within the family) distinguished

SA from every baseline tested, using only the seed alignment and no pretraining, backpropagation, or GPU resources. A systematic comparison with a Potts model fitted via pseudo-likelihood maximization (plmDCA) across all eight families (SI Appendix, Table S6) provided a more informative contrast, because Potts models are the most natural physics-based competitor: they explicitly fit pairwise coupling parameters $J_{ij}(a, b)$ between all position pairs, giving them direct access to coevolutionary signal. The comparison revealed comparable compositional fidelity and broadly similar output quality in six of eight families, with SA achieving higher novelty in seven of eight. However, the Potts model showed clear failure modes in the most data-starved families (Pkinase: novelty of 0.22 vs. 0.48 for SA, indicating the Gibbs sampler remained trapped near stored patterns; Defensin_beta: KL=0.112, the worst of any method), consistent with its extreme over-parameterization: the number of coupling parameters scales as $L(L-1)q^2/2$, yielding data-to-parameter ratios as low as 6.2×10^{-5} . That SA achieves comparable or better output without fitting any parameters suggests that the Hopfield energy implicitly captures the correlational structure that pairwise coevolutionary models fit explicitly, but through the softmax attention operation rather than through parameter estimation. Whereas the Potts model encodes pairwise couplings, the softmax nonlinearity in the Hopfield energy couples all positions simultaneously through the attention weights, effectively encoding all-order correlations without the combinatorial explosion of explicit higher-order terms.

Two control experiments confirmed that SA captured more than marginal per-position amino acid frequencies (SI Appendix, Tables S11 and S12). First, a consensus-with-noise baseline that added calibrated isotropic Gaussian perturbations in PCA space produced a qualitatively different output profile: systematically higher novelty (0.63–0.79 vs. 0.40–0.65 for SA) paired with lower sequence identity to stored patterns (0.44–0.62 vs. 0.52–0.66), reflecting the fact that isotropic noise scattered sequences away from all stored patterns rather than interpolating between them in a structured way. Second, SA run on column-permuted alignments, which preserved the exact per-position amino acid frequencies but destroyed all inter-position correlations, yielded consistently different output: higher novelty and lower sequence identity than SA on the real alignment in all eight families, with MI correlation differences favoring the real alignment in seven of eight families. The larger differences in novelty and sequence identity confirm that inter-position correlations present in real alignments constrain the energy landscape in ways that marginal frequencies alone cannot reproduce, and that SA exploits this correlational structure during generation.

The computational cost of SA differed from learned baselines by orders of magnitude (SI Appendix, Table S10). SA generated 150 sequences in 0.2–4.6 seconds across all eight families on a single CPU core, with the variation driven primarily by the number of stored patterns K (which determines the cost of the softmax attention at each Langevin step). Profile HMMs were consistently the fastest method (0.02–0.09 seconds), as expected for sampling from a precomputed position-specific model. EvoDiff required 2–14 hours on CPU depending on sequence length, and the MSA Transformer required 1–8 hours for iterative masked language model sampling. Potts model fitting and Gibbs sampling required 1–52 minutes, with the Pkinase family ($L=262$) dominating due to the $\mathcal{O}(L^2q^2)$ parameter count. These comparisons confirm that SA occupies a qualitatively different computational regime from learned baselines: it is 3–5 orders of magnitude faster than EvoDiff and the MSA Transformer, requires no GPU, no pretraining data, and no parameter optimization, and can be run on a laptop in the time it takes to open a web browser. This accessibility is not incidental; it is a direct consequence of the closed-form score function, which eliminates the need for backpropagation, gradient accumulation, or iterative parameter updates.

We assessed structural plausibility by predicting folded structures for 50 sequences per method per family using ESMFold (Fig. 4), comparing SA generation to stored natural sequences via Wilcoxon rank-sum tests with Cohen’s d effect sizes. SA-generated sequences achieved mean predicted local distance difference test (pLDDT) scores statistically indistinguishable from stored sequences in six of eight families ($p > 0.05$), with significantly higher pLDDT in Kunitz (90.8 vs. 89.3, $p=0.001$, $d=0.67$), and across all families at least 86% of SA-generated sequences exceeded the pLDDT=70 confidence threshold. Template modeling (TM) scores showed a consistent pattern: in six of eight families, SA generation achieved significantly higher TM-scores than stored sequences ($p < 0.05$; $d > 1.0$ in RRM, WW, PDZ, and Pkinase), likely reflecting the Boltzmann

ensemble's tendency to weight the central tendency of the family distribution, producing sequences closer to a consensus fold than individual natural sequences, which carry lineage-specific structural deviations. As an independent cross-validation, we repeated the structure prediction for all eight families using AlphaFold2 (SI Appendix, Figs. S3–S4). The AF2 results corroborated the ESMFold findings while resolving the single negative result: for SH3, the only family where ESMFold had indicated a pLDDT deficit, AlphaFold2 showed no pLDDT difference ($p=0.99$) and significantly higher TM-scores for SA generation (0.736 vs. 0.721, $p < 0.001$). The concordance between two independent structure predictors across all eight families strengthens the computational evidence that SA-generated sequences are predicted to adopt the correct fold (Fig. 5).

Pairwise mutual information (MI) analysis tested whether SA preserved residue covariation, which reflects structural contacts and functional couplings that single-site statistics cannot capture (SI Appendix, Table S3; Figs. S6–S7). Pearson correlations between stored and SA-generated MI vectors ranged from $r=0.53$ to 0.92 across seven of eight families, with Pkinase ($r=0.05$) a consistent outlier for all methods owing to its extreme data-to-position ratio ($K=37$, $L=262$). SA consistently exceeded profile HMMs ($r=0.07$ – 0.83), and the general ordering $\text{HMM} < \text{SA} < \text{Potts} \leq \text{EvoDiff}$ held across most families, consistent with the information hierarchy of each approach: position-independent frequencies, implicit all-order Hopfield couplings, explicitly fitted pairwise couplings, and deep coevolutionary priors from large-scale pretraining, respectively. This ordering confirmed that the Hopfield energy captured correlational structure beyond what position-independent models encode, without requiring explicit pairwise parameter estimation.

As a complementary, sequence-level assessment of biological plausibility, we scored all SA-generated and stored sequences using ESM2-650M (22), a protein language model pretrained on millions of UniRef50 sequences that is entirely independent of the stochastic attention framework (SI Appendix, Table S4). SA-generated sequences were statistically indistinguishable from natural family members in four of eight families (RRM, zf-C2H2, Pkinase, Defensin_beta; Wilcoxon $p > 0.29$), with SA achieving significantly *better* pseudo-perplexity in the Kunitz domain (4.47 vs. 4.98, $p < 0.001$, Cohen's $d=-0.74$). Three families showed modestly higher pseudo-perplexity for SA generation (SH3, WW, PDZ; $d=0.60$ – 1.03), though all SA-generated sequences remained well within the range of natural proteins. Across all eight families, the mean pseudo-perplexity was 5.71 for SA-generated sequences versus 5.58 for stored sequences, confirming that an independent model trained on billions of natural sequences considers SA-generated sequences comparably protein-like to natural family members.

We further cross-referenced SA-generated substitutions against published deep mutational scanning (DMS) datasets to ask whether the Boltzmann sampling procedure preferentially generates experimentally functional mutations (SI Appendix, Tables S4 and S5). For the PDZ domain (DLG4/PSD-95 PDZ3 (23)), 91.7% of 4,313 matched substitutions were classified as experimentally tolerated, compared to an overall tolerance rate of 77.5% in the DMS dataset ($1.18\times$ enrichment, binomial $p < 10^{-135}$). For the SH3 domain (GRB2 SH3 (24)), the enrichment was even stronger: 90.4% of 2,396 matched substitutions were tolerated versus 67.5% overall ($1.34\times$ enrichment, binomial $p < 10^{-156}$). In both families, the continuous DMS fitness scores of SA-generated substitutions were significantly higher than those of random single mutations (Wilcoxon $p < 10^{-21}$; SI Appendix, Tables S4 and S5). These results demonstrate that the Boltzmann sampling procedure preferentially generates substitutions at positions and to residues that are experimentally functional, without any explicit fitness objective.

Finally, the beta defensin family (PF00711) provided an opportunity to test whether SA generation preserves not only sequence-level statistics but also the biophysical properties that underpin antimicrobial function (SI Appendix, Fig. S8). SA-generated sequences preserved the six-cysteine disulfide scaffold (6.01 ± 0.08 mean cysteines, compared to 6.02 ± 0.15 for natural sequences), maintained a cationic charge profile ($+6.1 \pm 1.6$) with the narrowest charge distribution of any method, and occupied the same region of charge-hydrophobicity space as natural defensins. By a minimal antimicrobial plausibility filter (net charge $\geq +2$, hydrophobic fraction in $[0.3, 0.7]$, ≥ 4 cysteines), 51% of SA-generated sequences passed, compared to 42% of stored sequences, 25% of HMM-emitted sequences, and 16% of MSA Transformer sequences. This

implicit enforcement of biophysical constraints arises because the energy landscape concentrates probability near the stored patterns, which already satisfy these constraints by virtue of being functional proteins; no explicit objective for charge, hydrophobicity, or disulfide topology is required.

Discussion

We set out to ask what can be generated from a small protein family alignment with no training. The answer: sequences that are novel, compositionally faithful, and predicted by independent structure predictors to adopt the correct three-dimensional fold, produced in seconds on a laptop from nothing but the alignment itself. Across eight Pfam families spanning a ten-fold range of family sizes ($K=37-420$) and a seven-fold range of sequence lengths ($L=23-262$), stochastic attention produced output quality comparable to methods pretrained on millions of sequences, while requiring zero external data, zero training, and zero GPU resources. This combination of properties was not achieved by any single learned baseline we tested. Profile HMMs and the MSA Transformer generate sequences with sequence identity well below the natural within-family range (30–70%), indicating that high novelty scores for these methods reflect compositional drift rather than meaningful exploration. EvoDiff requires millions of pretraining sequences and hours of compute, and collapses on the longest family. No baseline simultaneously occupied the within-family identity range, maintained low composition divergence, and produced genuine novelty; stochastic attention is the only method that does. The fact that a training-free method, one that treats the seed alignment as both the model and the data, produces output comparable to systems built on orders of magnitude more information suggests that the statistical structure of protein families is rich enough to support generation without learned representations, provided the generative framework respects the geometry of the family's sequence space. The consensus-with-noise and column-permuted alignment controls confirmed that this is not a trivial consequence of marginal statistics: SA produces a qualitatively different output profile from isotropic perturbation of the consensus, and exploits inter-position correlations that column-independent models cannot capture.

The structure validation results were noteworthy. SA-generated sequences achieved significantly higher TM-scores to the canonical family fold in six of eight families by ESMFold, and this finding was corroborated by AlphaFold2 across all eight families, with TM-score concordance between the two predictors reaching $r=0.999$ (SI Appendix, Fig. S4). The pLDDT confidence scores were statistically indistinguishable from stored sequences in six of eight families, confirming that the sampler does not sacrifice folding confidence for diversity, and the single exception (SH3) was resolved by AlphaFold2 cross-validation. Two explanations for the elevated TM-scores merit consideration. The first is predictor bias: structure predictors are trained on evolutionary data, so they may assign higher confidence to consensus-proximal sequences that resemble the central tendency of their training distribution. The second is genuine structural convergence: consensus-proximal sequences may adopt a fold closer to the canonical family structure because the Boltzmann ensemble averages over lineage-specific structural deviations, producing sequences that encode the shared fold more cleanly than any individual natural sequence. These explanations are not mutually exclusive, and distinguishing them would require experimental structure determination. The DMS validation provides partial evidence for the latter interpretation: if the elevated TM-scores reflected only predictor bias rather than genuine foldability, one would not expect the generated substitutions to be enriched for experimentally tolerated mutations, yet the $1.18-1.34\times$ enrichment observed in two independent DMS datasets suggests that the sequences do encode functional proteins. We note, however, that all computational validators used in this study (structure predictors, ESM2-650M, and DMS-derived fitness scores) are trained on evolutionary data, so high scores for family-like sequences are expected; experimental structure determination and functional characterization of selected generated sequences would provide definitive confirmation.

The scaling study on the WW domain showed that stochastic attention is not fragile: generation quality was stable down to $K=20$ sequences, with KL divergence below 0.02 and novelty above 0.47 even at this extreme, and replication across three additional families confirmed generality. This robustness arises from a structural property of the method: the energy landscape is defined entirely by the stored patterns, with no

parameters that could overfit to a small training set. There is no encoder to memorize, no decoder to collapse, and no denoising network to undertrain. The entropy inflection criterion automatically adapts the operating temperature to the geometry of the memory matrix at each scale, and the critical temperature β^* itself can be predicted from PCA dimensionality alone ($R^2=0.97$), eliminating the need for a temperature sweep and enabling fully automatic operation from a seed alignment. The $\sqrt{d}$ scaling of β^* is not merely empirical: it follows from concentration of measure on the unit sphere, which guarantees that cross-similarities between stored patterns have variance $1/d$, so the inverse temperature required to resolve individual patterns must grow as $\sqrt{d}$ to overcome this concentration. The reduced coefficient (0.28 vs. the ~ 1.6 predicted for random patterns) reflects self-similarity anchoring: because each stored pattern has unit inner product with itself but only $\mathcal{O}(1/\sqrt{d})$ similarity to other patterns, the softmax concentrates on the self-pattern at much lower β than a mean-field model predicts (SI Appendix, Section S18). In contrast, learned generative models face a fundamental parameter-to-data ratio problem when $K < 100$: VAEs and diffusion models have thousands to millions of parameters that cannot be meaningfully constrained by a few dozen sequences. This regime encompasses most Pfam families and the long tail of protein space where generative tools are most needed.

Looking beyond proteins, the connection between attention and Boltzmann sampling suggests a broader principle: any set of examples, regardless of domain, defines an energy landscape via the modern Hopfield energy, and Langevin dynamics on this landscape provides a principled generative model with no training. The requirements are minimal: the examples must be representable as vectors in a common space, and the practitioner must choose an inverse temperature that balances fidelity and diversity. Two deeper conditions determine whether the approach will produce useful output. First, the stored examples must be structured enough that the Hopfield energy landscape has well-defined basins, each corresponding to a cluster or prototype in the data, rather than a featureless high-dimensional plateau; protein families satisfy this condition because evolutionary constraint creates correlated patterns of conservation and variation. Second, the examples must be diverse enough that the sampler is not trivially retrieval: if all stored patterns are nearly identical, the Boltzmann distribution collapses to a point mass and generation reduces to consensus with noise. The eight families studied here span a range of within-family diversity (22–37% mean pairwise identity), confirming that the method operates in the intermediate regime between these extremes. It is worth noting the historical irony: classical Hopfield networks were criticized for limited storage capacity, storing at most $\mathcal{O}(d)$ patterns before catastrophic interference. The modern continuous Hopfield energy, combined with Langevin sampling, turns this limitation into a feature for scarce-data generation: with $K \ll d$ stored patterns, the energy landscape is well-separated and the sampler can explore the space between memories without interference, precisely the regime where deep generative models fail. This perspective may extend naturally to other biological sequence families where data is scarce but examples are structured, such as antibody repertoires where individual clonotypes may have only a handful of somatic variants, RNA families where Rfam seed alignments are typically smaller than their Pfam counterparts, or small-molecule libraries where chemical series may contain only tens of analogs. Indeed, the Hopfield energy admits a natural conditioning mechanism: adjusting pattern multiplicities via a scalar bias on the attention logits continuously shifts generation from the full family toward a user-specified functional subset, enabling property-directed generation (e.g., toward binding activity) without retraining (25). The analytic score function also provides a natural interface with the convergence theory of Langevin dynamics (26, 27), enabling formal guarantees on sample quality that are absent from most neural generative models and opening a path toward certifiable generation in safety-critical applications such as therapeutic protein design.

From a practical standpoint, the computational accessibility of stochastic attention may be its most consequential property. The entire pipeline, from seed alignment to generated sequences, is implemented in Julia using only standard numerical libraries (LinearAlgebra, Statistics, MultivariateStats), requires no GPU, no pretraining data, and no hyperparameter tuning beyond the automatically determined temperature. Generation of 150 sequences completes in under 5 seconds on a consumer laptop for all eight families tested (SI Appendix, Table S10). This stands in contrast to the infrastructure requirements of deep generative

models: EvoDiff requires access to the UniRef50 database and GPU resources for pretraining; the MSA Transformer requires a pretrained 100M-parameter model and iterative masked inference; even Potts model fitting requires L-BFGS-B optimization over hundreds of thousands of coupling parameters. For the majority of protein families, which are small and studied by individual laboratories without dedicated computational infrastructure, these requirements place deep generative methods out of reach. Stochastic attention removes this constraint: the method requires only a seed alignment and a standard workstation, and produces sequences that preserve family statistics and are predicted to fold correctly.

Several limitations should be acknowledged. The most significant is that stochastic attention operates on a fixed alignment representation: it requires a pre-computed MSA and a fixed alignment length L , and cannot generate insertions, deletions, or variable-length sequences. This constraint is inherited from the one-hot-plus-PCA encoding, which maps each sequence to a vector of fixed dimension $20L$; any change in L would alter the representation space itself. In practice, this means the method is best suited to compact, well-aligned domains rather than multi-domain proteins or families with frequent large insertions. Extending stochastic attention to variable-length generation would likely require replacing PCA with a length-agnostic embedding while preserving the closed-form score function, a nontrivial but tractable direction for future work. The PCA projection itself discards information that may be relevant for capturing higher-order correlations, though a sensitivity analysis confirmed robustness across the 75%–99% retained variance range (SI Appendix, Table S13). The eight families studied here are all well-structured, globular domains with abundant structural and functional annotation; this choice was necessary for validation but introduced a selection bias toward precisely the class of proteins that structure predictors handle best. Performance on intrinsically disordered regions, transmembrane domains, multi-domain proteins, or families with very low sequence identity remains untested, and the method’s reliance on PCA may be particularly limiting for disordered families where sequence variation is less structured. Finally, our multi-chain protocol does not guarantee coverage of all modes in the Boltzmann distribution, particularly for very large or highly diverse families, and wet-lab characterization of selected generated sequences would be needed to establish the method’s utility for protein engineering applications.

Materials and Methods

Stochastic Attention. Let $\mathbf{X} = [\mathbf{m}_1, \dots, \mathbf{m}_K] \in \mathbb{R}^{d \times K}$ be the memory matrix whose columns are vectorized protein sequences from a family alignment of K members, each represented in d dimensions after one-hot encoding and PCA projection. Each sequence of length L is first one-hot encoded to a binary vector in $\{0, 1\}^{20L}$, then projected by PCA (retaining 95% of variance, yielding $d \ll 20L$) and unit-normalized to form the corresponding column $\mathbf{m}_k$; full encoding, projection, and decoding details are given in SI Appendix, Section S17. The modern Hopfield energy (19) is:

$$E(\boldsymbol{\xi}) = \frac{1}{2} \|\boldsymbol{\xi}\|^2 - \frac{1}{\beta} \log \sum_{k=1}^K \exp(\beta \mathbf{m}_k^\top \boldsymbol{\xi}), \quad [1]$$

where $\beta > 0$ is an inverse temperature. The Boltzmann distribution $p_\beta(\boldsymbol{\xi}) \propto \exp(-\beta E(\boldsymbol{\xi}))$ has score function $\nabla \log p_\beta(\boldsymbol{\xi}) = \beta(\mathbf{T}(\boldsymbol{\xi}) - \boldsymbol{\xi})$, where $\mathbf{T}(\boldsymbol{\xi}) := \mathbf{X} \text{softmax}(\beta \mathbf{X}^\top \boldsymbol{\xi})$ is the softmax attention retrieval map. This score is *exact* and computed by a single attention operation; no score network or training is required. Applying the unadjusted Langevin algorithm (ULA) (28) to the potential $U = \beta E$ yields the stochastic attention update:

$$\boldsymbol{\xi}_{t+1} = (1 - \alpha) \boldsymbol{\xi}_t + \alpha \mathbf{X} \text{softmax}(\beta \mathbf{X}^\top \boldsymbol{\xi}_t) + \sqrt{\frac{2\alpha}{\beta}} \boldsymbol{\epsilon}_t, \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad [2]$$

Each step performs contraction toward the origin, a softmax-weighted pull toward stored memories, and isotropic Gaussian noise, at a per-step cost of $\mathcal{O}(dK)$.

Temperature Selection and Sequence Decoding. The critical inverse temperature β^* is identified via the entropy inflection criterion: we evaluate the Boltzmann entropy $H(\beta) = -\sum_k w_k(\beta) \log w_k(\beta)$, where $w_k(\beta) = \text{softmax}(\beta \mathbf{X}^\top \boldsymbol{\xi})_k$ are the attention weights at a representative stored pattern $\boldsymbol{\xi} = \mathbf{m}_1$, and locate the inflection point $\beta^* = \arg \max_\beta |d^2 H / d\beta^2|$ via a grid search over $\beta \in [0.1, 20]$. Below β^* the sampler visits all patterns equiprobably;

Algorithm 1 Stochastic Attention Protein Sequence Generation

Require: Seed alignment $\mathbf{S} \in \mathcal{A}^{K \times L}$, variance threshold $\rho_{\min} = 0.95$, step size $\alpha = 0.01$, chains $N_c = 30$, iterations $T = 5000$, burn-in $T_b = 2000$, thinning $\Delta t = 100$, samples/chain $s = 5$

Ensure: Generated sequences $\{\mathbf{g}_1, \dots, \mathbf{g}_S\} \subset \mathcal{A}^L$, $S = N_c \cdot s$

```
1: Encoding:
2: One-hot encode:  $\mathbf{X} \leftarrow [\text{onehot}(\mathbf{S}_{1,:}), \dots, \text{onehot}(\mathbf{S}_{K,:})] \in \{0, 1\}^{20L \times K}$ 
3: Center:  $\bar{\mathbf{x}} \leftarrow \frac{1}{K} \sum_k \mathbf{X}_{:,k}$ ;  $\tilde{\mathbf{X}} \leftarrow \mathbf{X} - \bar{\mathbf{x}} \mathbf{1}_K^\top$ 
4: SVD:  $\tilde{\mathbf{X}} = \mathbf{U} \Sigma \mathbf{V}^\top$ 
5: Select  $d$ : smallest  $p$  such that  $\sum_{j=1}^p \sigma_j^2 / \sum_j \sigma_j^2 \geq \rho_{\min}$ 
6: Project and normalize:  $\mathbf{m}_k \leftarrow \mathbf{W}_d^\top (\mathbf{X}_{:,k} - \bar{\mathbf{x}}) / \|\mathbf{W}_d^\top (\mathbf{X}_{:,k} - \bar{\mathbf{x}})\|$  for  $k = 1, \dots, K$ 
7: Assemble memory:  $\hat{\mathbf{X}} \leftarrow [\mathbf{m}_1, \dots, \mathbf{m}_K] \in \mathbb{R}^{d \times K}$ 
8: Temperature selection:
9: Find  $\beta^*$ : entropy inflection of  $H(\beta) = -\sum_k p_k \log p_k$  with  $p_k = \text{softmax}(\beta \hat{\mathbf{X}}^\top \mathbf{m}_k)_k$ 
10: Set  $\beta_{\text{gen}} \leftarrow \max(\lceil 2\beta^* \rceil, 5)$ 
11: Multi-chain Langevin sampling:
12:  $\mathcal{G} \leftarrow \emptyset$ 
13: for  $c = 1, \dots, N_c$  do
14:   Draw  $k_c \sim \text{Uniform}\{1, \dots, K\}$ 
15:    $\xi_0 \leftarrow \mathbf{m}_{k_c} + 0.01 \boldsymbol{\eta}$ ,  $\boldsymbol{\eta} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$ 
16:   for  $t = 0, \dots, T - 1$  do
17:      $\mathbf{a}_t \leftarrow \text{softmax}(\beta_{\text{gen}} \hat{\mathbf{X}}^\top \xi_t)$ 
18:      $\boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$ 
19:      $\xi_{t+1} \leftarrow (1 - \alpha) \xi_t + \alpha \hat{\mathbf{X}} \mathbf{a}_t + \sqrt{2\alpha/\beta_{\text{gen}}} \boldsymbol{\epsilon}_t$ 
20:   Collect  $s$  evenly spaced samples from  $\{\xi_{T_b+1}, \xi_{T_b+\Delta t+1}, \dots, \xi_T\}$  into  $\mathcal{G}$ 
21: Decoding:
22: for each  $\xi \in \mathcal{G}$  do
23:   Inverse PCA:  $\hat{\mathbf{x}} \leftarrow \bar{\mathbf{x}} + \mathbf{W}_d \xi \in \mathbb{R}^{20L}$ 
24:   Argmax decode:  $g_\ell \leftarrow \arg \max_{a \in \mathcal{A}} \hat{x}_{20(\ell-1)+a}$  for  $\ell = 1, \dots, L$ 
25: return  $\{\mathbf{g}_1, \dots, \mathbf{g}_S\}$ 
```

above β^* pattern-specific basins emerge. Generation uses $\beta_{\text{gen}} \approx 2\beta^*$, just above the inflection, to balance exploration and fidelity.

To decode a PCA-space sample ξ to an amino acid sequence, the PCA projection is inverted to recover a one-hot-like vector in the original $20L$ -dimensional space, and the amino acid at each position is assigned by argmax over the 20 channels. The resulting length- L sequence is the decoded output; positions where the PCA reconstruction assigns negligible weight to all amino acids default to the most frequent residue in the stored alignment at that position. The complete pipeline is summarized in Algorithm 1.

Experimental Design. We selected eight Pfam families spanning a range of sizes, sequence lengths, and conservation levels (Table 1): RRM (PF00076, $K=68$, $L=71$), SH3 (PF00018, $K=55$, $L=48$), WW (PF00397, $K=420$, $L=31$), Kunitz (PF00014, $K=99$, $L=53$), zf-C2H2 (PF00096, $K=151$, $L=23$), PDZ (PF00595, $K=44$, $L=83$), Pkinase (PF00069, $K=37$, $L=262$), and Defensin_beta (PF00711, $K=45$, $L=36$). Seed alignments were downloaded from InterPro (18) and cleaned by removing columns with $> 50\%$ gaps and sequences with $> 30\%$ gaps. PCA retained 95% of variance, followed by unit-norm projection to form the memory matrix. For each family, we ran 30 independent ULA chains of $T=5,000$ iterations ($\alpha=0.01$), discarding 2,000 burn-in iterations and thinning every 100 steps to retain 5 samples per chain ($S=150$ sequences total). The inverse temperature was set automatically via the entropy inflection criterion, with a generation regime at $\beta_{\text{gen}} \approx 2\beta^*$ and a retrieval regime at $\beta_{\text{ret}} \approx 20\beta^*$. The thinning interval of 100 steps is well-matched to the integrated autocorrelation time of the Hopfield energy ($\tau_{\text{int}} = 76\text{--}122$ steps across families), yielding approximately 35 effective independent samples per chain and $\approx 1,050$ effective draws

total; the burn-in of 2,000 steps provides a $3.2\text{--}6.6\times$ margin over the empirical chain convergence point (SI Appendix, Table S10). Initialization sensitivity was verified by repeating the full pipeline with random sphere initialization ($\xi_0 = \mathbf{z}/\|\mathbf{z}\|$, $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$) in place of stored-pattern initialization; output metrics were indistinguishable across all eight families (maximum absolute KL difference 0.0003), confirming that the generated ensemble reflects the Boltzmann distribution rather than an artifact of starting conditions (SI Appendix, Table S11).

To contextualize SA against methods that require pretraining on large external corpora, we compared four established baselines: (i) profile HMM emit via HMMER3 (5); (ii) EvoDiff MSA-conditioned diffusion (15), pretrained on UniRef50 MSAs; (iii) MSA Transformer (14) via iterative Gibbs-like masked language model sampling; and (iv) a Potts model fitted via pseudo-likelihood maximization (plmDCA). For each baseline, the same 150-sequence budget and evaluation metrics (KL divergence, novelty, sequence identity) were applied. Structural plausibility was assessed by predicting folded structures for 50 sequences per method per family using ESMFold (22) and AlphaFold2 (1) (via ColabFold (29)), comparing each predicted structure to an experimentally determined reference using TM-align (30). All statistical comparisons used Wilcoxon rank-sum tests with Benjamini-Hochberg false discovery rate (FDR) correction at $q=0.05$.

Acknowledgments. The idea for stochastic attention arose while developing a lecture on energy-based methods for eCornell. We thank the Pfam/InterPro consortium for maintaining publicly available seed alignments. Structure predictions were performed using ESMFold and AlphaFold2 (via ColabFold). Computations were performed on resources at Cornell University.

Data, Materials, and Software Availability

All code, data, seed alignments, generated sequences, and analysis scripts are publicly available at <https://github.com/varnerlab/SA-Protein-Modeling-Study> (21). Pfam seed alignments were obtained from InterPro (18). No new experimental data were generated for this study; all validation used publicly available computational tools (ESMFold, AlphaFold2 via ColabFold, ESM2-650M) and published deep mutational scanning datasets.

1. Jumper J, et al. (2021) Highly accurate protein structure prediction with AlphaFold. *Nature* 596:583–589.
2. Baek M, et al. (2021) Accurate prediction of protein structures and interactions using a three-track neural network. *Science* 373(6557):871–876.
3. Kuhlman B, Bradley P (2019) Advances in protein structure prediction and design. *Nature Reviews Molecular Cell Biology* 20:681–697.
4. Huang PS, Boyken SE, Baker D (2016) The coming of age of de novo protein design. *Nature* 537(7620):320–327.
5. Potter SC, et al. (2018) HMMER web server: 2018 update. *Nucleic Acids Research* 46(W1):W200–W204.
6. Krogh A, Brown M, Mian IS, Sjölander K, Haussler D (1994) Hidden Markov models in computational biology: Applications to protein modeling. *Journal of Molecular Biology* 235(5):1501–1531.
7. Morcos F, et al. (2011) Direct-coupling analysis of residue coevolution captures native contacts across many protein families. *Proceedings of the National Academy of Sciences* 108(49):E1293–E1301.
8. Ekeberg M, Lökvist C, Lan Y, Weigt M, Aurell E (2013) Improved contact prediction in proteins: Using pseudolikelihoods to infer Potts models. *Physical Review E* 87(1):012707.
9. Trinquier J, Uguzzoni G, Pagnani A, Zamponi F, Weigt M (2021) Efficient generative modeling of protein sequences using simple autoregressive models. *Nature Communications* 12:5800.
10. Kingma DP, Welling M (2014) Auto-encoding variational Bayes in *International Conference on Learning Representations (ICLR)*.
11. Riesselman AJ, Ingraham JB, Marks DS (2018) Deep generative models of genetic variation capture the effects of mutations. *Nature Methods* 15:816–822.
12. Ferruz N, Schmidt S, Höcker B (2022) ProtGPT2 is a deep unsupervised language model for protein design. *Nature Communications* 13:4348.
13. Madani A, et al. (2023) Large language models generate functional protein sequences across diverse families. *Nature Biotechnology* 41:1099–1106.
14. Rao R, et al. (2021) MSA transformer in *Proceedings of the 38th International Conference on Machine Learning (ICML)*. pp. 8844–8856.
15. Alamdari S, et al. (2023) Protein generation with evolutionary diffusion: Sequence is all you need. *bioRxiv*.
16. Dauparas J, et al. (2022) Robust deep learning-based protein sequence design using ProteinMPNN. *Science* 378(6615):49–56.
17. Watson JL, et al. (2023) De novo design of protein structure and function with RFdiffusion. *Nature* 620(7976):1089–1100.
18. Mistry J, et al. (2021) Pfam: The protein families database in 2021. *Nucleic Acids Research* 49(D1):D412–D419.
19. Ramsauer H, et al. (2021) Hopfield networks is all you need in *International Conference on Learning Representations (ICLR)*.
20. Krotov D, Hopfield JJ (2016) Dense associative memory for pattern recognition in *Advances in Neural Information Processing Systems*. Vol. 29.
21. Alswaidan A, Varner JD (2026) Stochastic attention via Langevin dynamics on the modern Hopfield energy. *arXiv preprint arXiv:2603.06875*.
22. Lin Z, et al. (2023) Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science* 379(6637):1123–1130.
23. McLaughlin RN, Poelwijk FJ, Raman A, Gosal WS, Ranganathan R (2012) The spatial architecture of protein function and adaptation. *Nature* 491:138–142.
24. Faure AJ, et al. (2022) Mapping the energetic and allosteric landscapes of protein binding domains. *Nature* 604:175–183.
25. Varner JD (2026) Conditioning protein generation via Hopfield pattern multiplicity. *arXiv preprint arXiv:2603.20115*.
26. Durmus A, Moulines É (2017) Nonasymptotic convergence analysis for the unadjusted Langevin algorithm. *The Annals of Applied Probability* 27(3):1551–1587.
27. Dalalyan AS (2017) Theoretical guarantees for approximate sampling from smooth and log-concave densities. *Journal of the Royal Statistical Society: Series B* 79(3):651–676.
28. Roberts GO, Tweedie RL (1996) Exponential convergence of Langevin distributions and their discrete approximations. *Bernoulli* 2(4):341–363.
29. Mirdita M, et al. (2022) ColabFold: Making protein folding accessible to all. *Nature Methods* 19:679–682.
30. Zhang Y, Skolnick J (2005) TM-align: a protein structure alignment algorithm based on the TM-score. *Nucleic Acids Research* 33(7):2302–2309.

Significance Statement

Most protein families have fewer than 100 known members, too few to train a deep generative model. We show that a single attention operation over a family alignment defines an energy landscape whose Boltzmann distribution can be sampled via Langevin dynamics, with no training, no GPU, and no external data. The resulting method, stochastic attention, generates novel protein sequences that preserve family-level amino acid statistics and are predicted by independent structure predictors to adopt the correct fold across eight diverse families. A simple linear relationship between representation dimensionality and the critical sampling temperature enables fully automatic operation from a seed alignment, opening training-free sequence generation to the majority of protein families too small for deep learning.

J.D.V. designed research, performed research, analyzed data, and wrote the paper.

The author declares no competing interest.

¹ To whom correspondence should be addressed. E-mail: jdv27@cornell.edu. ORCID: 0000-0002-2558-7026.

Table 1. Properties of the eight Pfam families used in this study. K : number of seed sequences after cleaning. L : alignment length. d : PCA dimension (95% variance). $\bar{H}_{\text{col}}$: mean per-column Shannon entropy (nats). K_{eff} : effective number of sequences at 80% identity. β^* : empirical entropy inflection point.

Family	Pfam ID	K	L	d	$\bar{H}_{\text{col}}$	K_{eff}	β^*	$\beta^* / \sqrt{d}$
RRM	PF00076	68	71	59	1.92	68	3.85	0.50
SH3	PF00018	55	48	46	1.68	55	3.85	0.57
WW	PF00397	420	31	186	1.74	419	5.45	0.40
Kunitz	PF00014	99	53	80	1.61	99	3.85	0.43
zf-C2H2	PF00096	151	23	106	1.91	151	4.58	0.44
PDZ	PF00595	44	83	37	1.88	43	3.23	0.53
Pkinase	PF00069	37	262	34	1.72	37	3.23	0.55
Defensin_beta	PF00711	45	36	37	1.65	45	3.23	0.53

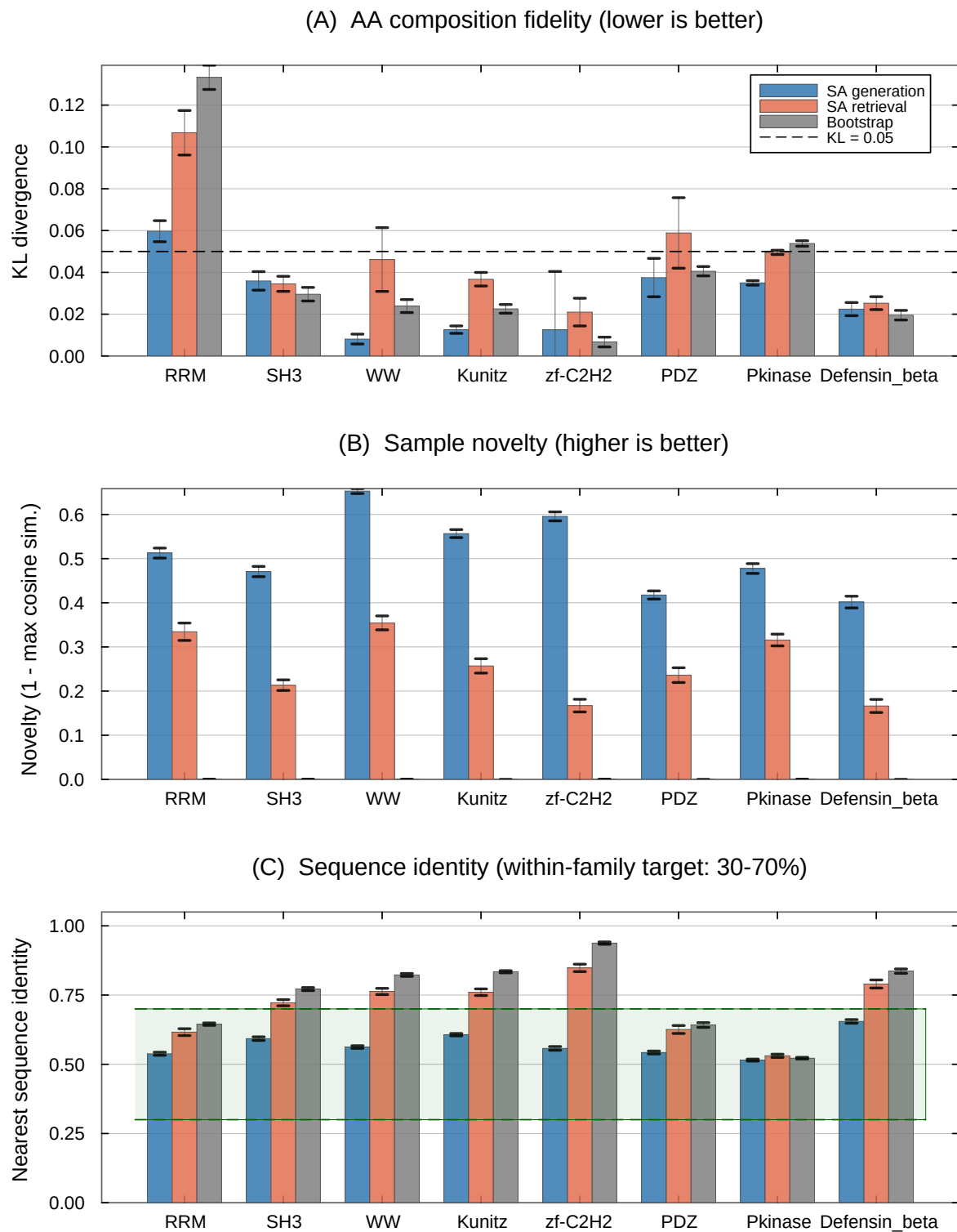


Fig. 1. Cross-family comparison of SA generation, SA retrieval, and bootstrap across eight Pfam families. **(A)** KL divergence of amino acid composition (lower is better); dashed line at $KL=0.05$. **(B)** Novelty in PCA space ($1 - \max_k \cos(\hat{\xi}, \mathbf{m}_k)$; higher is better). Bootstrap samples have zero novelty by construction. **(C)** Nearest sequence identity to a stored pattern; green shaded band (30–70%) marks the typical within-family identity range. Error bars: ± 1 SE (30 chains $\times$ 5 samples).

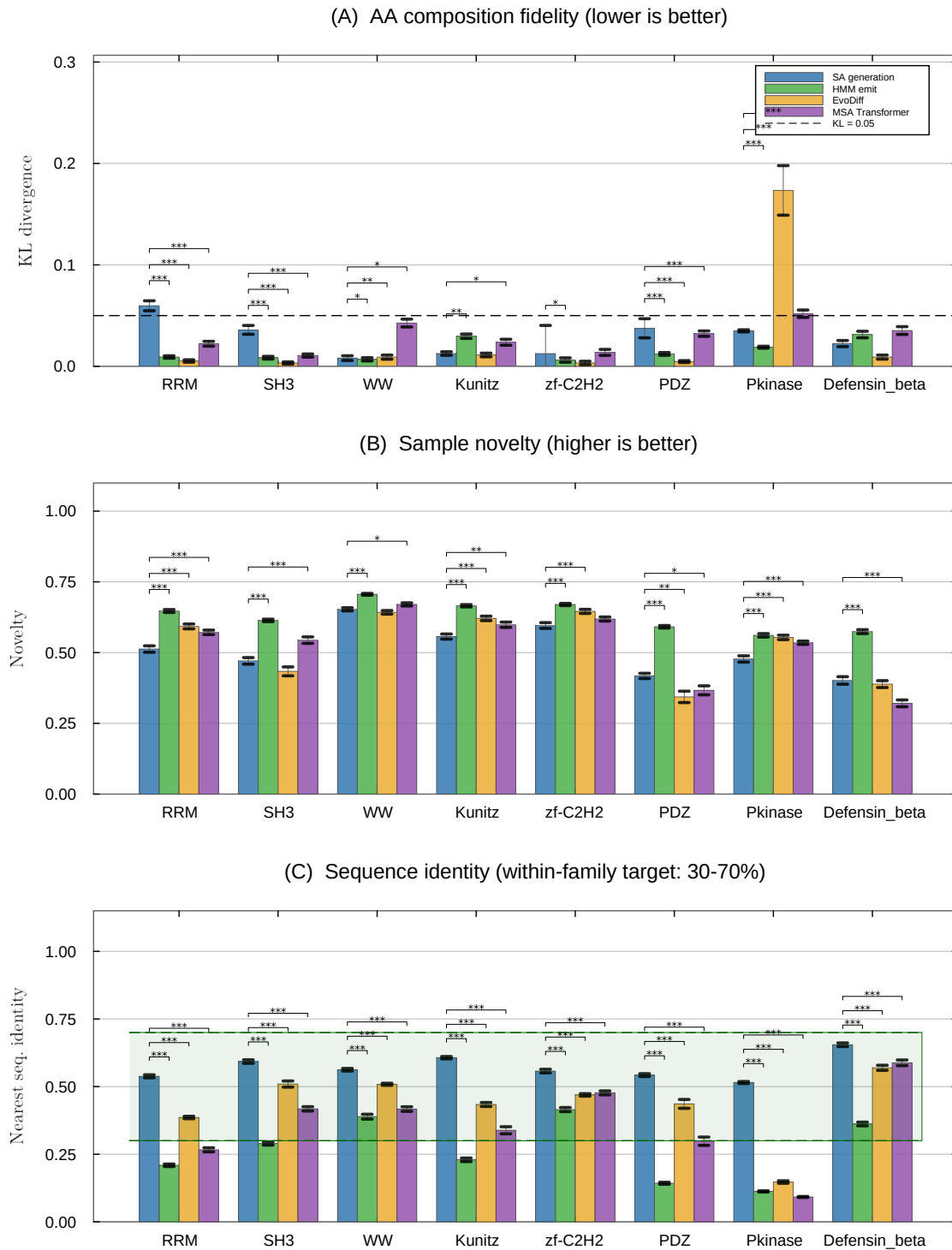


Fig. 3. Head-to-head comparison of SA generation against three learned baselines across all eight Pfam families. **(A)** KL divergence of amino acid composition (bootstrap SE; $n_{\text{boot}}=1000$); dashed line at $\text{KL}=0.05$. **(B)** Novelty in PCA space. **(C)** Nearest sequence identity to a stored pattern; green shaded band (30–70%) marks the within-family identity range. Panels (B) and (C): per-chain means ± 1 SE. Significance brackets: Wilcoxon rank-sum tests ($***p < 0.001$, $**p < 0.01$, $*p < 0.05$). Reference baselines are reported in SI Appendix, Table S1.

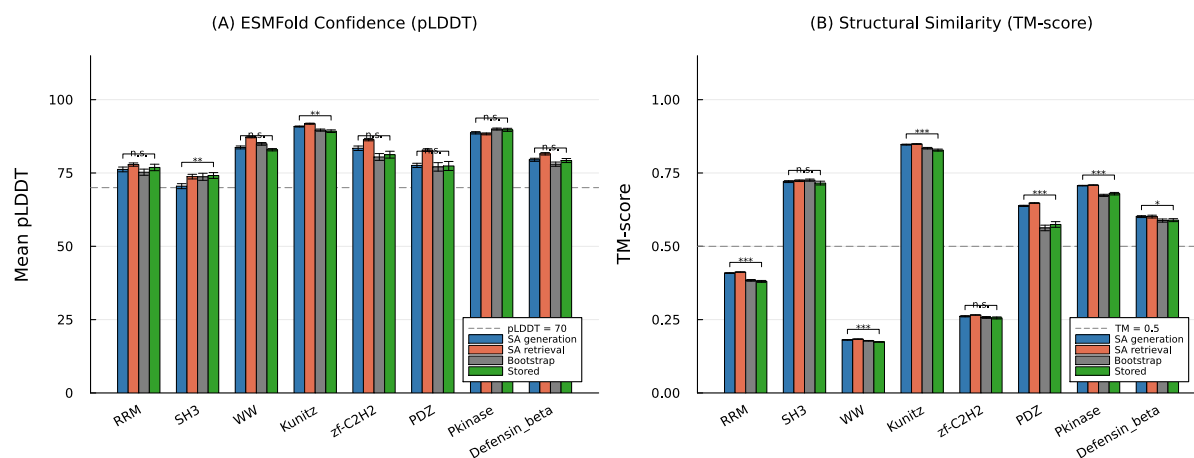


Fig. 4. Structure validation using ESMFold. **(A)** Mean pLDDT across all eight families for SA generation, SA retrieval, bootstrap, and stored sequences; dashed line at pLDDT=70. **(B)** TM-score to the experimentally determined reference structure; dashed line at TM=0.5. Significance: Wilcoxon rank-sum tests (*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, n.s. = not significant). Error bars: ± 1 SE ($n=50$).

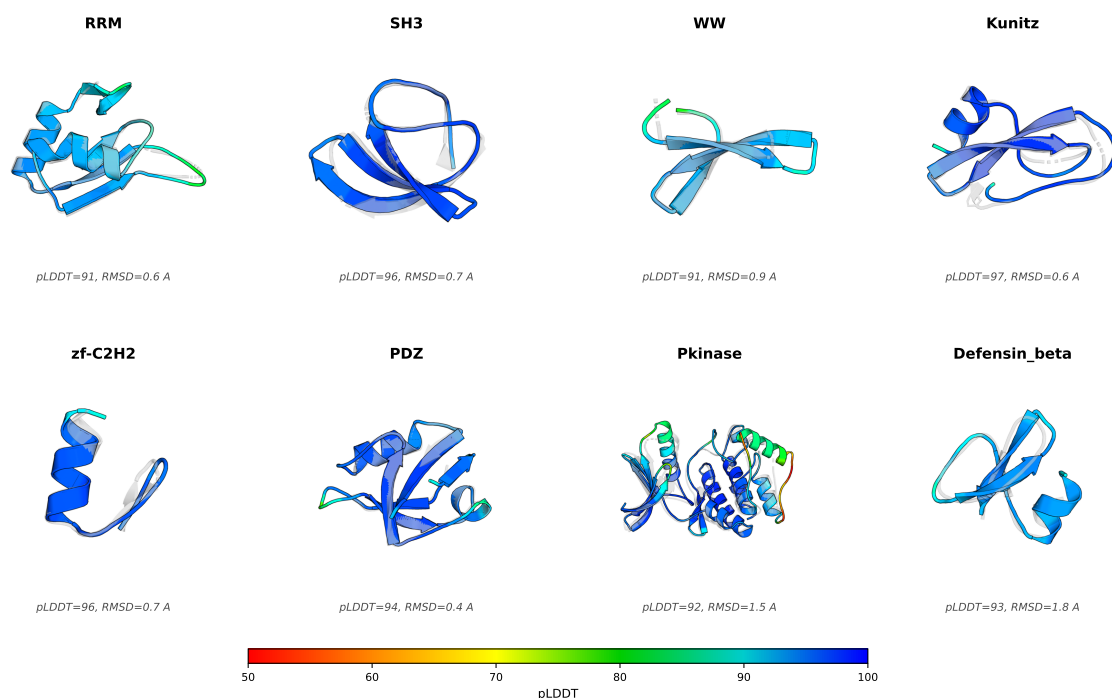


Fig. 5. AlphaFold2 predicted structures for representative SA-generated sequences (colored by per-residue pLDDT confidence) superimposed on the experimentally determined reference structure for each family (gray, semi-transparent). For each family, the SA-generated sequence with the highest mean pLDDT was selected and aligned to the reference using PyMOL. The close predicted structural agreement (sub-angstrom root-mean-square deviation (RMSD)) and uniformly high pLDDT scores are consistent with SA-generated sequences encoding three-dimensional folds that recapitulate the canonical architecture of their target family, despite 40–65% sequence novelty.